Supporting Scientific Sovereignty:

Evaluating the Feasibility of Partner-Led Respiratory Pathogen Surveillance in Global Health Alliances

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KEY FINDING

The partner-led site in Choma, Zambia outperformed the university-managed site in Madison, WI on both testing consistency (87.5% vs. 46.9%) and turnaround time (2.0 vs. 3.0 days; p < 0.001)

1. INTRODUCTION

Global health surveillance historically relies on extractive research models where data is collected locally but controlled by external institutions, undermining local sustainability and scientific agency.

This project co-created a decentralized, partner-led pathogen surveillance model in Choma, Zambia, operated by Macha Research Trust — an independent Zambian research institution with over two decades of experience in infectious disease research in Southern Province.

RESEARCH QUESTION

Can a decentralized, partner-led diagnostic model achieve the same longitudinal data integrity as a directly managed domestic model?

2. CONCEPTUAL FRAMEWORK

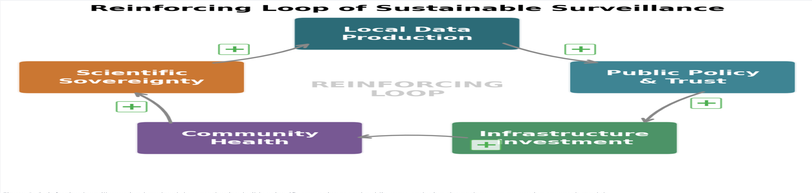

Figure 1: Reinforcing loop illustrating how local data production builds scientific sovereignty and public trust, reducing dependence on extractive research models.

3. METHODS

Using Cepheid GeneXpert testing, we quantified Influenza A/B, RSV, and SARS-CoV-2 from air samplers deployed in healthcare settings at both sites (July 2025 – March 2026).

- Choma site: InBio Apollo air sampler, operated by Macha Research Trust staff
- Madison site: ThermoFisher AerosolSense, operated by UW-Madison lab team
- Target: 4 machines × 3 samples/week = 12 samples/week per site
- Metrics: testing frequency, turnaround time (sampling start to PCR result)

Note:
Different air samplers were used at each site. Operational metrics (TaT, frequency) are comparable; pathogen yields may differ.

4. RESULTS

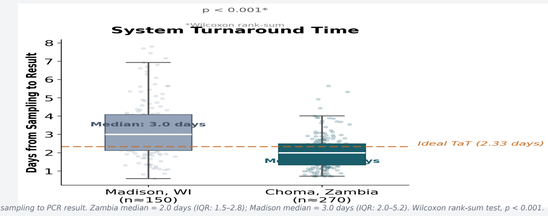

Figure 2: Turnaround time from sampling to PCR result. Zambia median = 2.0 days (IQR: 1.5–2.8); Madison median = 3.0 days (IQR: 2.0–5.2). Wilcoxon rank-sum test, p < 0.001.

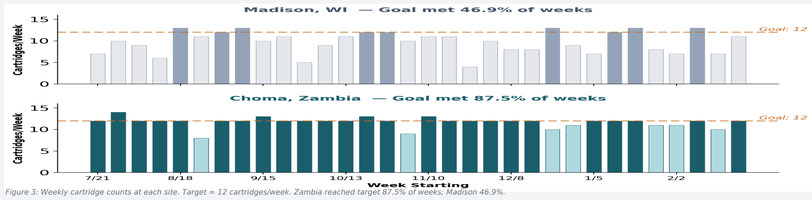

Figure 3: Weekly cartridge counts at each site. Target = 12 cartridges/week. Zambia reached target 87.5% of weeks; Madison 46.9%.

5. THEORY OF CHANGE

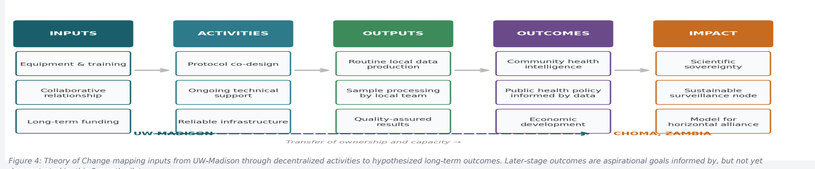

Figure 4: Theory of Change mapping inputs from UW-Madison through decentralized activities to hypothesized long-term outcomes. Later-stage outcomes are aspirational goals informed by, but not yet demonstrated in, this 8-month pilot.

6. DISCUSSION

The Choma site's superior performance on both metrics challenges the assumption that directly managed sites produce more reliable data.

Madison underperformance reflects healthcare staffing constraints and competing lab priorities that created collection lag — a challenge less acute at the Choma site where surveillance is a primary mandate for the dedicated Macha Research Trust team.

The Choma site maintained its 2.0-day median turnaround despite operational challenges including insects entering sampling equipment, demonstrating adaptability under real-world conditions.

Limitations

- Eight-month observation window captures one respiratory season; multi-year data needed
- Different air samplers at each site may affect pathogen yield comparisons
- Cost-per-site and long-term financial sustainability not yet assessed

7. CONCLUSION

Decentralized, partner-led surveillance models can match and exceed the operational efficiency of directly managed sites. The Madison–Choma partnership demonstrates that horizontal alliances built on scientific sovereignty provide a viable foundation for non-extractive global health infrastructure.

Next steps include multi-season validation, pathogen circulation analysis across sites, cost-effectiveness evaluation, and assessment of data utilization by local public health authorities in Southern Province.

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